

SERVICENOW

MEDICAL DIAGNOSTIC CENTER

Project Report

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ServiceNow Global Certification Project

Patient Management | Appointment Scheduling | Report Generation

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1. INTRODUCTION

The Medical Diagnostic Center application built on ServiceNow automates and streamlines patient registration, diagnostic test catalog management, appointment booking, report generation, email notifications, and health analytics. In a modern healthcare environment, manual record-keeping leads to delays in diagnostic reporting, scheduling conflicts, and communication gaps between healthcare providers and patients.

This project demonstrates a custom ServiceNow application designed to manage the entire lifecycle of a patient's interaction with a diagnostic laboratory. From cataloging laboratory tests (such as Blood Tests, COVID-19 PCR, and Imaging) to generating diagnostic reports and dispatching automated email notifications, the solution provides an end-to-end, traceable, and secure workflow.

Project Overview

- **Platform:** ServiceNow Custom Application Development
- **Primary Module:** Medical Diagnostic Management
- **Supporting Modules:** Platform Analytics, Email Notification Engine, System Logs
- **Primary Roles:** Patient / End-User, Lab Technician, Medical Approver / Admin, System Administrator
- **Main Activities:** Test catalog definition, patient onboarding, appointment booking, automated billing & approval routing, report generation, email notification triggering, and reporting dashboard analytics
- **Key Entities:** Patient Table, Diagnostic Test Table, Appointment Table, Report Table, Email Notifications, Health Reports Dashboard
- **Outcome:** A centralized, automated, and secure healthcare diagnostic management portal
- **Overall Benefit:** Faster report turnaround times, automated patient notifications, zero double-booking, transparent billing, and enhanced operational visibility via analytics dashboards

2. PROJECT OBJECTIVES AND SCOPE

2.1 Business Objectives

- Standardize diagnostic test cataloging with dynamic pricing, durations, and test codes.
- Streamline patient registration while securely storing personal health data.
- Provide a self-service portal for patients to search tests, schedule appointments, and receive confirmation notifications.
- Automate the approval workflow for test appointments, payment processing, and report delivery.
- Enable lab technicians to upload findings and automated email notifications with dynamic field scripts ({patient.first_name}, {report_number}).
- Deliver real-time analytics to clinic managers via custom Platform Analytics dashboards.

2.2 Functional Scope

- Custom table creation: Diagnostic Test, Patient, Appointment Table, and Report Table.
- Field-level configurations including reference fields, currency fields, choice dropdowns, HTML editors, and date-time pickers.
- Dynamic self-service portal UI for patient appointment scheduling and live confirmation tracking.
- Automated email notifications for appointment status and report availability.
- Custom Analytics Dashboard displaying diagnostic metrics by location and test category.

2.3 Stakeholders

Stakeholder	Role	Needs / Expectations	Automation Impact
Patients	End Users	Easy booking, transparent pricing, timely report delivery	Self-service portal booking, email notifications with dynamic report links
Lab Technicians	Operations	Structured form entry for findings, clear schedule visibility	Standardized report entry tables, auto-linked patient & appointment context
Medical Approvers	Quality Control	Verification of test results before patient dispatch	One-click approval workflows and status tracking
Clinic Managers	Operations / Management	Operational insights, patient throughput, test volumes	Interactive Platform Analytics dashboards
ServiceNow Admin	System Admin	Scalable data schema, automated workflows, reliable email dispatch	Maintainable custom application scope with full audit logging

3. REQUIREMENT ANALYSIS AND PLANNING

3.1 Execution Roadmap

1. Application Scope & Data Model Definition
2. Diagnostic Test Catalog Configuration
3. Patient Table Schema Setup
4. Appointment Portal & Processing Workflow Design
5. Report Generation Module Implementation
6. Automated Email Notification & Template Scripting
7. Platform Analytics Dashboard Creation
8. End-to-End System Integration Testing & Validation

3.2 System Architecture & Data Schema

The application relies on four primary custom tables built within the custom application scope (**x_1989081_medica_0**):

- **x_1989081_medica_0_diagnostic_test**: Stores test details, prices, codes, categories, and active status.
- **x_1989081_medica_0_patient**: Contains registered patient demographics and user references.
- **x_1989081_medica_0_appointment_table**: Links patients to diagnostic tests, tracking scheduled dates, slot times, payment status, approval status, and total fees.
- **x_1989081_medica_0_report_table**: Captures rich-text clinical findings, report files, verified-by references, and report numbers.

3.3 Expected Application Lifecycle

The patient selects a test from the portal catalog, picks a date and time slot, and submits the booking. The system creates an Appointment Record set to Pending Approval. Once approved by administrative staff, a confirmation email is automatically sent to the patient. Following test completion at the center, the lab technician opens the Report Table, enters clinical findings, attaches the test file, and submits it. Saving the report triggers an automated notification containing personalized details (`${patient.first_name}` and `${report_number}`) informing the patient that their report is ready for download.

4. DIAGNOSTIC TEST CATALOG & PATIENT MANAGEMENT

4.1 Diagnostic Test Configuration

The diagnostic test record defines individual lab tests available within the center. Fields include Test Name, Test Code, Category dropdown, Test Duration, Price (Currency), Active flag, and Description.

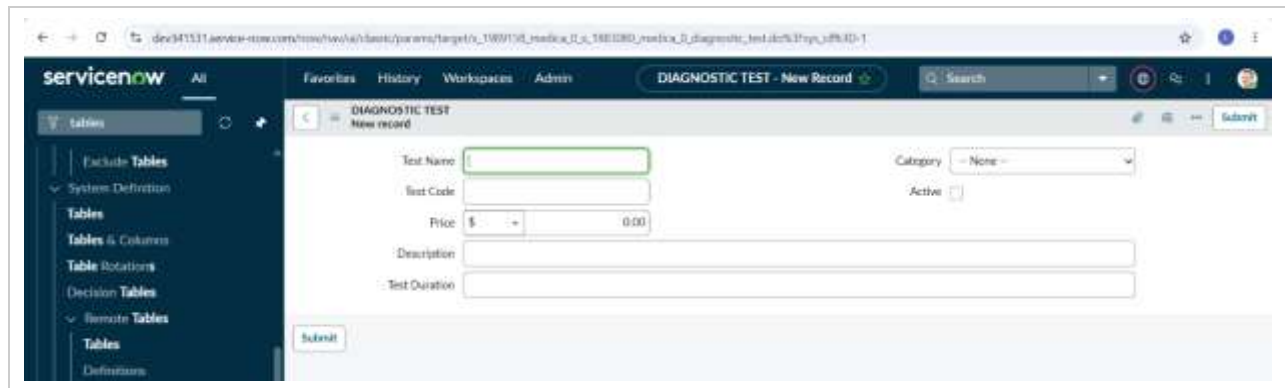


Figure 1: Diagnostic Test New Record Configuration Form

4.2 Diagnostic Test Inventory List

Once populated, the diagnostic test catalog provides a comprehensive view of all active procedures offered by the diagnostic center, including test pricing, codes, and duration.

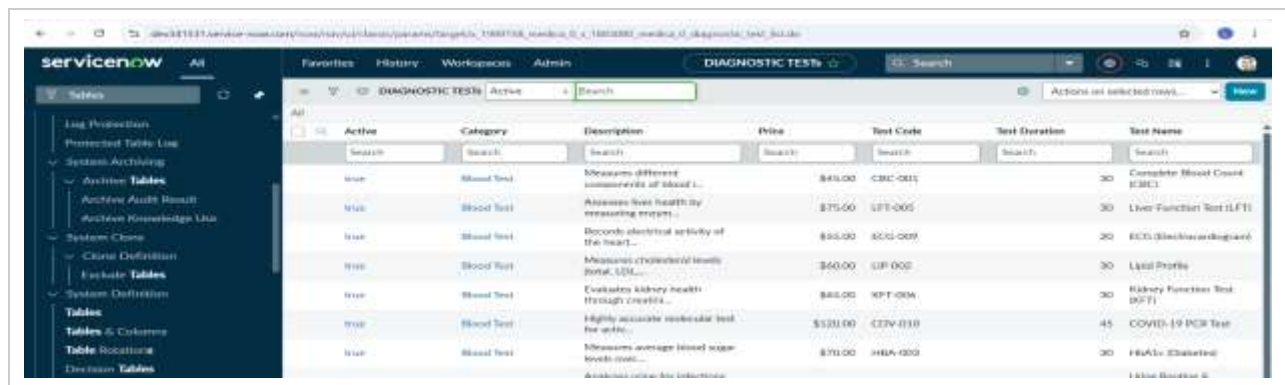


Figure 2: Active Diagnostic Test List View

Active	Category	Test Name	Test Code	Duration (Min)	Price
true	Blood Test	HbA1c (Diabetes)	HBA-003	30	\$70.00
true	Blood Test	Urine Routine & Microscopy	UR-007	20	\$25.00
true	Blood Test	Vitamin D Test	VIT-011	30	\$55.00
true	Blood Test	Complete Blood Count (CBC)	CBC-001	30	\$45.00

Active	Category	Test Name	Test Code	Duration (Min)	Price
true	Blood Test	Thyroid Function Test (TFT)	THY-004	30	\$95.00
true	Blood Test	Chest X-Ray	XRY-008	15	\$80.00
true	Blood Test	Iron Studies	IRN-012	30	\$50.00
true	Blood Test	Liver Function Test (LFT)	LFT-005	30	\$75.00
true	Blood Test	ECG (Electrocardiogram)	ECG-009	20	\$55.00
true	Blood Test	Lipid Profile	LIP-002	30	\$60.00
true	Blood Test	Kidney Function Test (KFT)	KFT-006	30	\$65.00
true	Blood Test	COVID-19 PCR Test	COV-010	45	\$120.00

4.3 Patient Registration & Data Management

Patient records are captured using the Patient form, which securely logs Patient ID, First Name, Last Name, Phone Number, Email, Gender (Female, Male, Other), Date Of Birth, Address, and the linked SysUser reference.

The screenshot shows the 'Patient - Create Created' form in the ServiceNow interface. The form is titled 'Patient - Create Created' and has a 'Submit' button. The form fields are as follows:

- Patient ID
- First Name
- Last Name
- Phone Number
- Email
- Gender: -- None -- (dropdown menu)
- Date Of Birth: -- None -- (dropdown menu)
- Address: Female (dropdown menu)
- User: Male (dropdown menu)

A 'Submit' button is located at the bottom of the form.

Figure 3: Patient Registration Form Configuration.

5. APPOINTMENT SCHEDULING & WORKFLOW MANAGEMENT

5.1 Booking & Scheduling Interface

Patients can select their required test, schedule a date/time slot, and submit the booking through a guided multi-step wizard. The portal homepage highlights available tests and average report turnaround time, giving patients a transparent, self-service entry point into the diagnostic center.

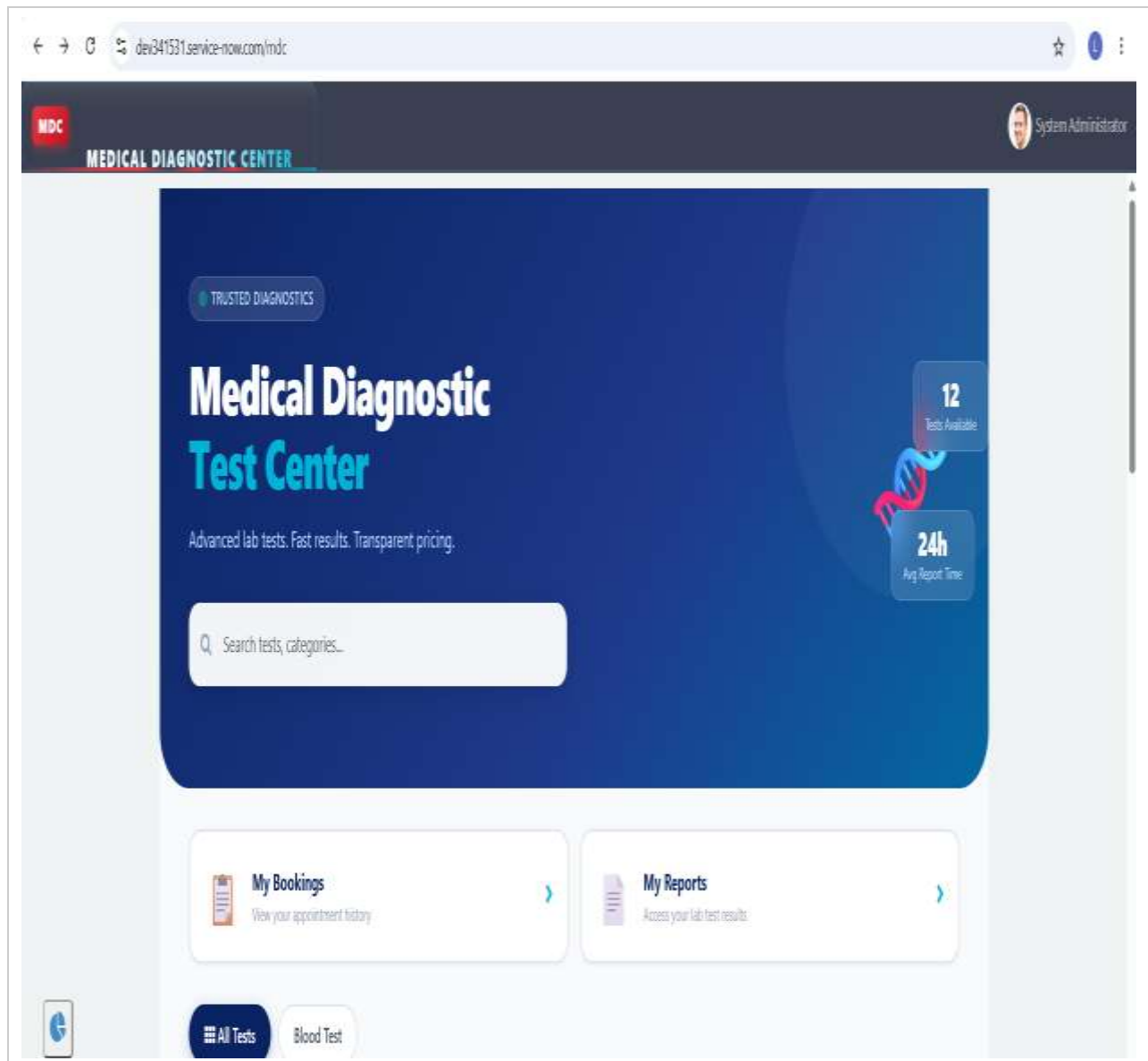


Figure 4: Self-Service Patient Portal Homepage

From the homepage, patients browse the full catalog of active diagnostic tests, filter by category, and initiate a booking directly against the desired test.

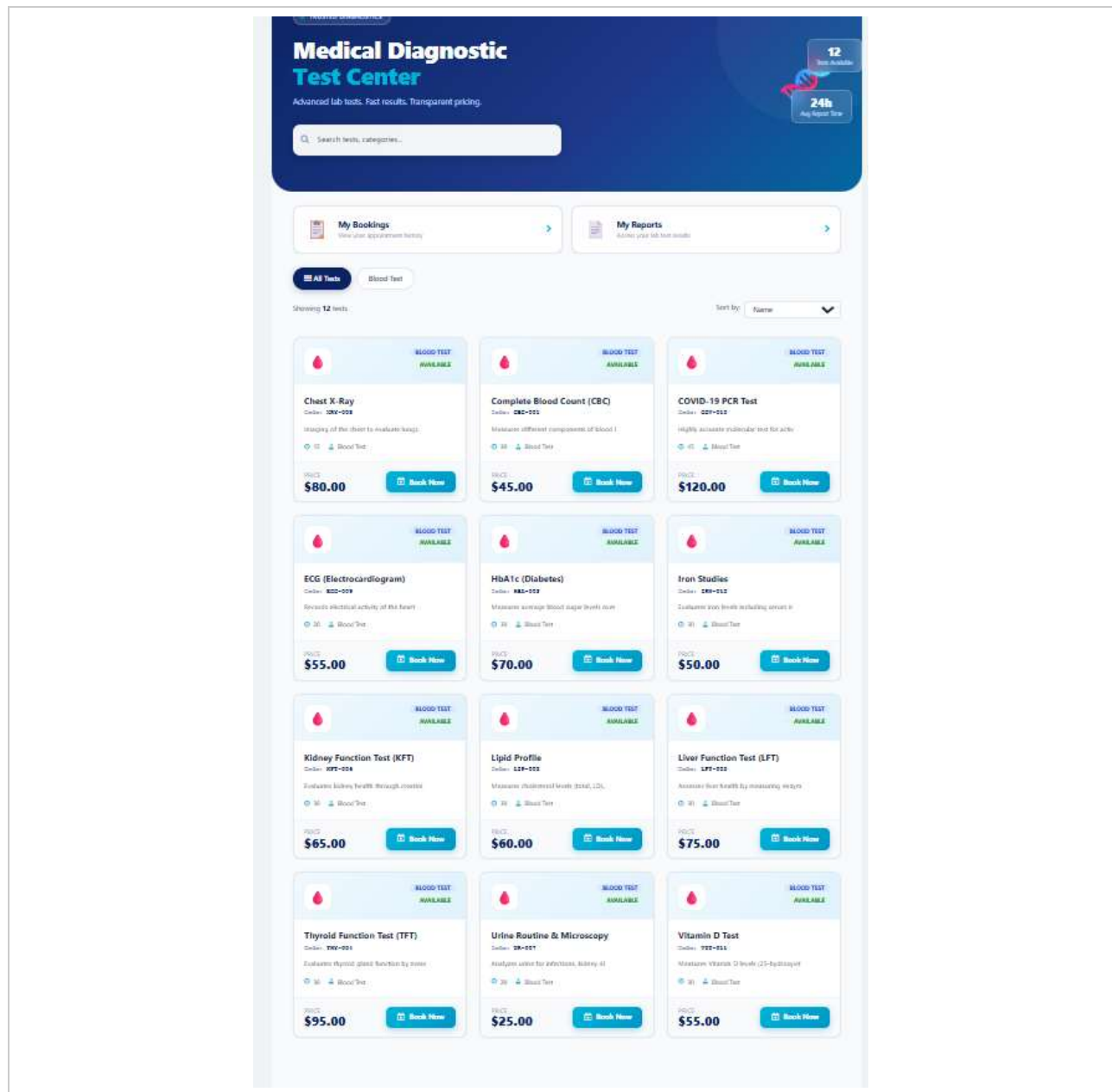


Figure 5: Diagnostic Test Catalog — Patient Portal View

Selecting a test opens the guided booking wizard. Step one collects the patient's personal information — name, contact details, date of birth, gender, and address — before the patient proceeds to schedule and pay for the appointment.

1

Details

2

Schedule

3

Payment

4

Done

 **Chest X-Ray**
Blood Test · Code: XRY-008 · Duration: 15

\$80.00
[Change Test](#)

 **Patient Information**
Fill in your personal details to proceed.

FIRST NAME *

System

LAST NAME *

Administrator

EMAIL ADDRESS *

admin@example.com

PHONE NUMBER *

+1 555 000 0000

DATE OF BIRTH *

dd-mm-yyyy

GENDER *

-- Select Gender --

ADDRESS

Your full address

[← Back to Catalog](#)

[Continue →](#)

Figure 6: Patient Information Entry — Booking Wizard, Step 1

Upon completing the scheduling and payment steps, the screen provides instant confirmation of the appointment ID (APPT-2026-24623), payment mode, total cost, and pending approval status.

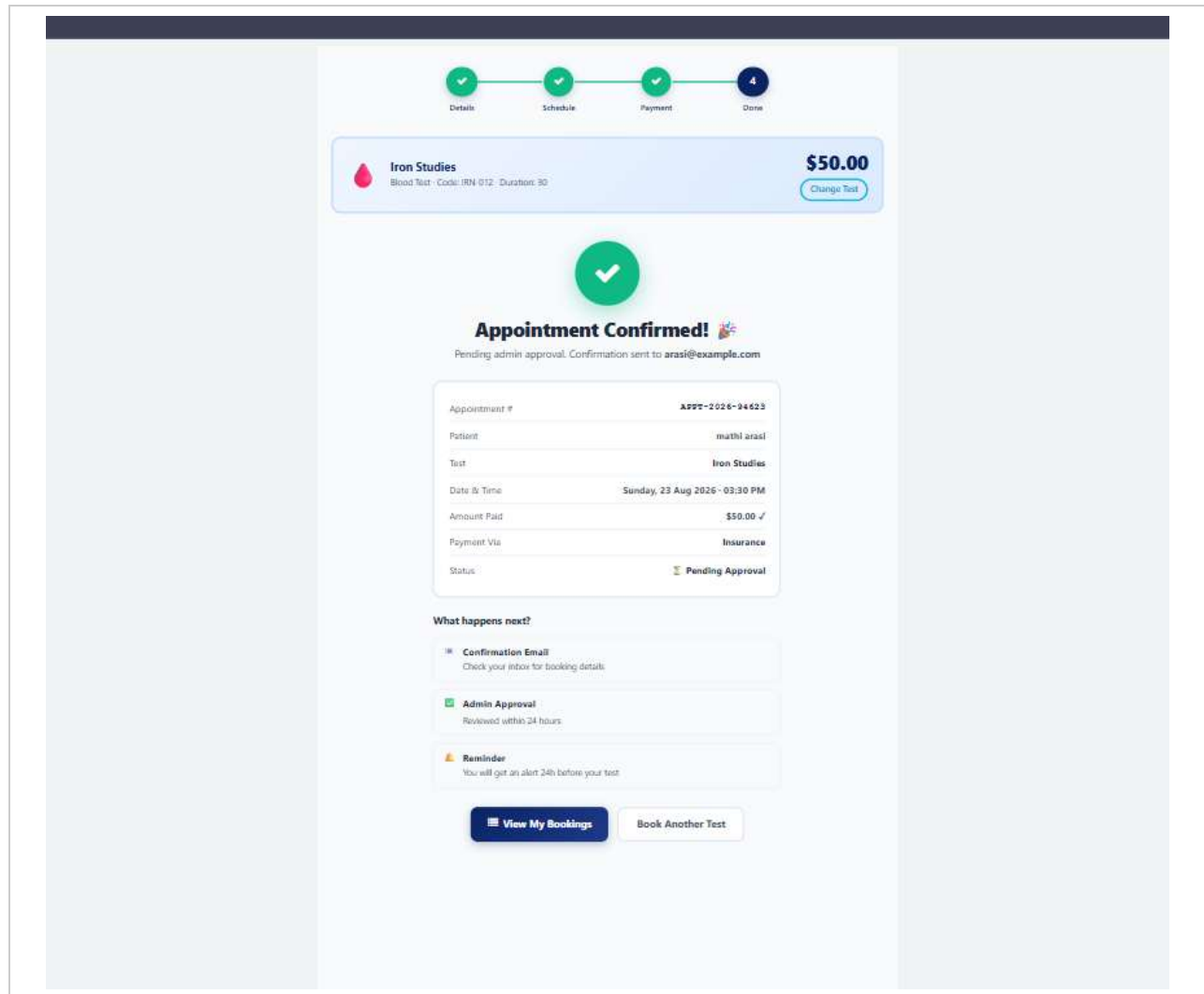


Figure 7: Patient Appointment Confirmation Interface

5.2 Appointment Table Schema & Record Structure

The back-end Appointment Table (x_1989081_medica_0_appointment_table) tracks all scheduled bookings. Administrators can view reference links to Patient and Diagnostic Test records, alongside financial and approval metadata.

Figure 8: Appointment Table Form Interface

The list view below shows the aggregate of appointment records, including approval status, confirmation flags, payment status, and the associated patient reference.

Appointment Number	Approval Date	Approval Status	Approved By	Confirmation Sent	Created By	Email	Notes	Patient	Payment Status
APPT-2026-15096	(empty)	requested	(empty)	false	System Administrator	info@example.com		x_189081_medica_0_patient	paid
Created 2026-08-25 06:33:15	2026-08-25 06:33:15	Approved	(empty)	false	(empty)	info@example.com		x_189081_medica_0_patient	Paid
Created 2026-08-25 06:33:15	(empty)	Completed	(empty)	false	(empty)			x_189081_medica_0_patient	
APPT-2026-88206	(empty)	requested	(empty)	true	System Administrator	admin@example.com		x_189081_medica_0_patient	paid
APPT-2026-16440	(empty)	Cancelled	(empty)	true	System Administrator	info@example.com		x_189081_medica_0_patient	paid

Figure 9: Appointment Table List View

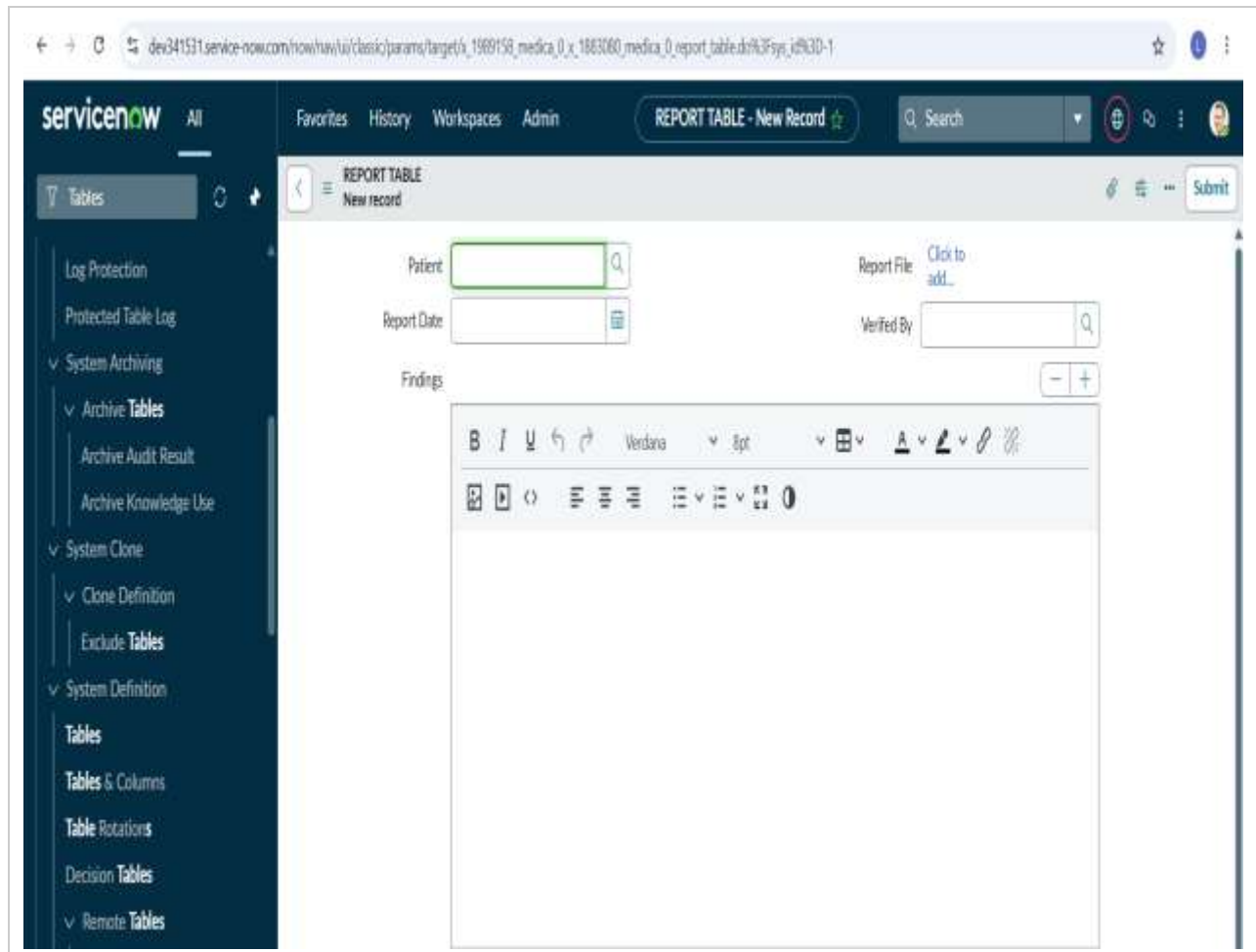
Appointment Record Field Map

- **Appointment Number:** Auto-generated unique string identifier (e.g., APPT-2026-24623).
- **Patient:** Reference field pointing to x_189081_medica_0_patient.
- **Status:** Choice dropdown (Pending, Confirmed, Completed, Cancelled).
- **Approval Status:** Choice dropdown (Pending Approval, Approved, Rejected).
- **Scheduled Date & Slot Time:** Date/Time picker capturing the exact slot reserved.
- **Test:** Reference field pointing to x_189081_medica_0_diagnostic_test.
- **Total Amount & Refund Amount:** Currency fields tracking financial transactions.
- **Payment Status:** Dropdown reflecting payment processing (Pending, Paid, Refunded).
- **Confirmation Sent:** Checkbox flag set automatically by notification scripts.

6. REPORT GENERATION & NOTIFICATION INTEGRATION

6.1 Diagnostic Report Schema

Upon completing a lab test, technicians record findings in the Report Table (x_1989081_medica_0_report_table). This table features an HTML editor for detailed clinical notes, file attachment fields for uploaded PDF lab results, and reference fields linking back to the specific Patient and Appointment.



The screenshot displays the ServiceNow 'REPORT TABLE - New Record' form. The interface includes a left-hand navigation menu with options like 'Log Protection', 'Protected Table Log', 'System Archiving', 'Archive Tables', 'Archive Audit Result', 'Archive Knowledge Use', 'System Clone', 'Clone Definition', 'Exclude Tables', 'System Definition', 'Tables', 'Tables & Columns', 'Table Rotations', 'Decision Tables', and 'Remote Tables'. The main form area contains several input fields: 'Patient' (with a search icon), 'Report Date' (with a calendar icon), 'Report File' (with a 'Click to add...' link), and 'Verified By' (with a search icon). Below these fields is a 'Findings' section featuring a rich text editor with a toolbar containing various formatting options (bold, italic, underline, link, unlink, text color, background color, bulleted list, numbered list, indent, outdent, undo, redo, and a '0' icon). The form is titled 'REPORT TABLE - New Record' and includes a 'Submit' button in the top right corner.

Figure 10: Diagnostic Report Form Interface

6.2 Automated Email Notification System

When a diagnostic report is verified and saved, ServiceNow triggers an automated email notification (reportsNotifications). The subject and body dynamically interpolate record attributes using script variables such as `${patient.first_name}` and `${report_number}`.

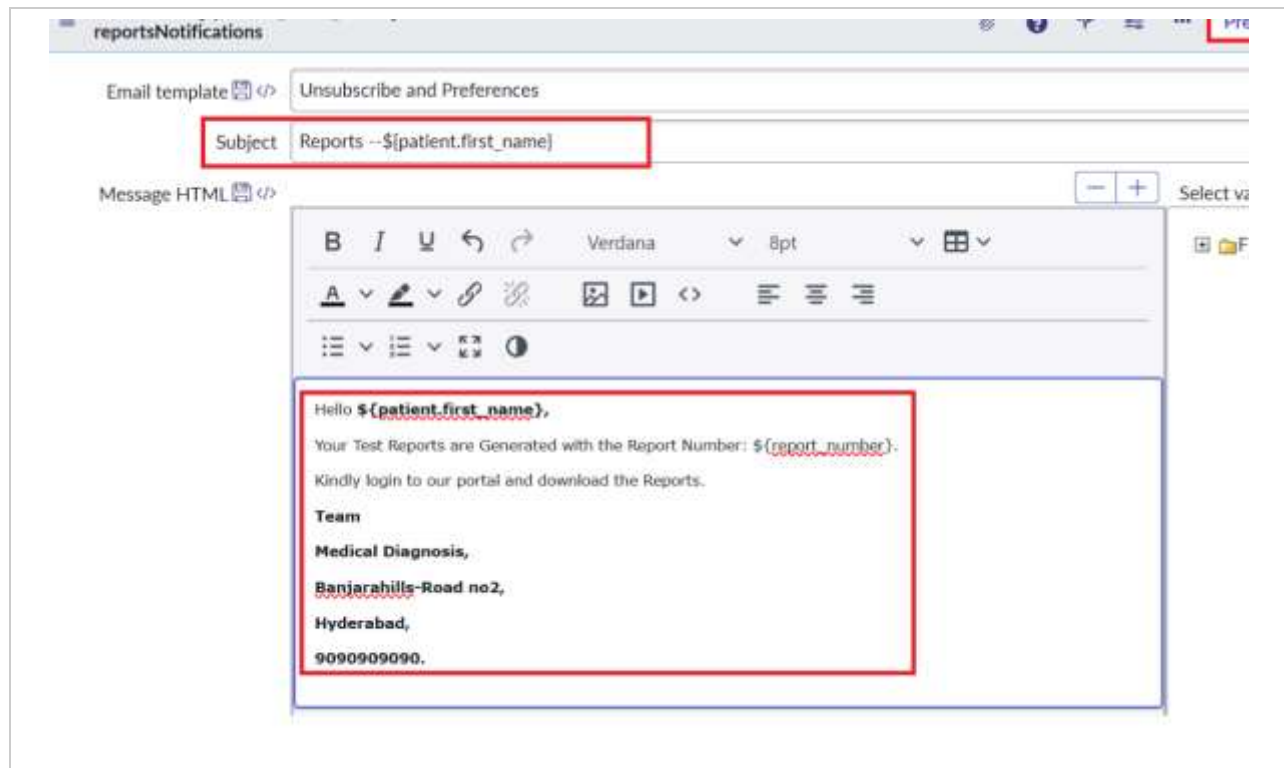


Figure 11: Report Email Notification Template Configuration

Email Notification Content

- **Subject:** Reports --\${patient.first_name}

Body:

Hello \${patient.first_name},

Your Test Reports are Generated with the Report Number: \${report_number}.

Kindly login to our portal and download the Reports.

Team

Medical Diagnosis,

Banjarahills-Road no2,

Hyderabad,

9090909090.

6.3 Email Dispatch Logs & Restocking Alerts

The system monitors outgoing mail using ServiceNow's native Email Engine (sys_email). System administrators can audit generated notification payloads, verify recipient addresses, review delivery statuses, and monitor system events (such as inventory restocking alerts for diagnostic supplies).

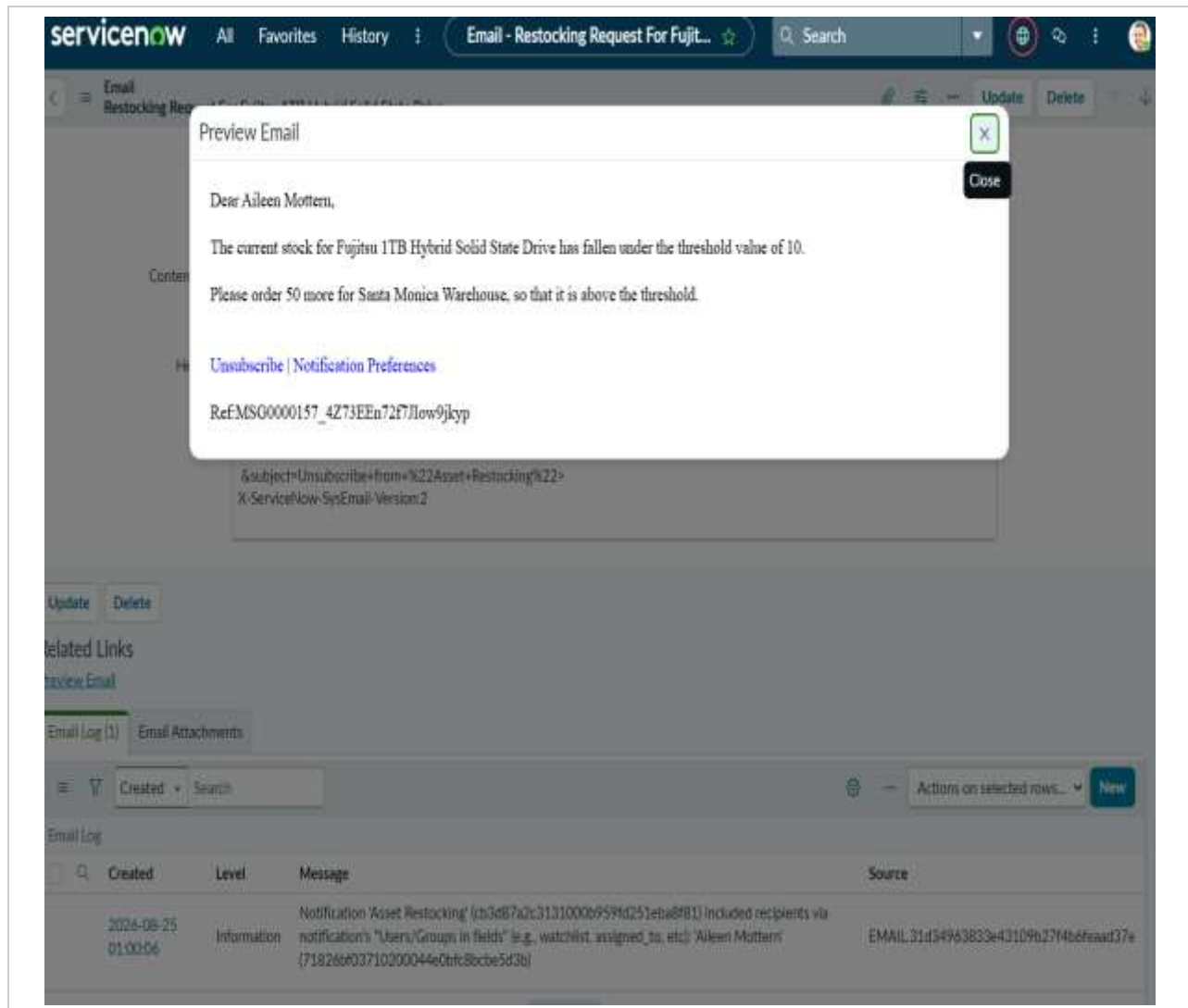


Figure 12: Email Preview and System Audit Log View

7. ANALYTICS & HEALTH REPORTS DASHBOARD

7.1 Platform Analytics & Reporting

To provide management with visibility into diagnostic center operations, a custom Health Reports Dashboard was configured using the ServiceNow Platform Analytics workspace.

7.2 Dashboard Visualization

The dashboard displays real-time metrics, such as report distribution across different regional branches/locations (e.g., Cuddalore vs. MYD centers).

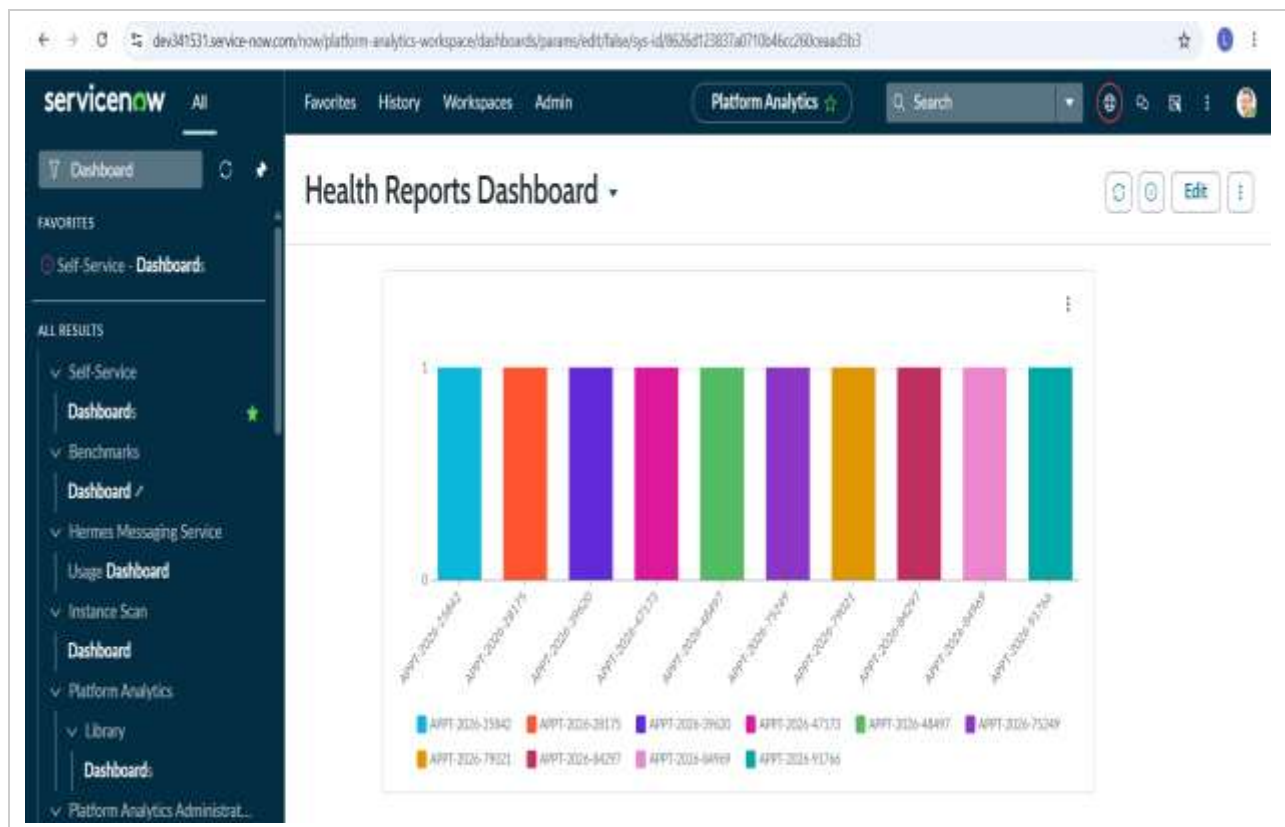


Figure 13: Health Reports Dashboard in Platform Analytics

8. FINAL VALIDATION, TESTING AND CONCLUSION

8.1 Final Validation

The final phase verifies that all operational steps — from patient registration to dashboard metrics — function seamlessly:

- **Patient Registration:** Successfully creates user profiles with linked contact details.
- **Test Catalog Access:** Tests are correctly listed with active pricing and descriptions.
- **Appointment Scheduling:** Appointments generate auto-numbered records and trigger approval workflows.
- **Report Generation:** Lab reports save HTML findings and attach diagnostic PDFs.
- **Email Notification Dispatch:** Emails render dynamic fields (`${patient.first_name}`, `${report_number}`) and log correctly in `sys_email`.
- **Platform Analytics:** Real-time dashboards accurately reflect location-based test counts.

8.2 End-to-End Testing & Deployment Validation

Phase	Main Activity	Expected Result	Status
1	Requirement Analysis & Data Modeling	Custom scoped tables defined (x_1989081_medica_0)	PASSED
2	Diagnostic Test Catalog Setup	Test catalog populated with active tests and prices	PASSED
3	Patient Registration	Patient demographic data accurately mapped	PASSED
4	Appointment Scheduling	Portal wizard schedules appointment with ID tracking	PASSED
5	Approval & Billing	Appointment status updates to pending/approved	PASSED
6	Report Generation	Clinical findings logged via rich-text editor	PASSED
7	Notification Triggering	Email generated with dynamic script interpolation	PASSED
8	Audit & Log Verification	Outgoing emails verified in ServiceNow Email Logs	PASSED
9	Analytics Dashboard	Real-time health report charts active in workspace	PASSED

8.3 Project Outcome

The Medical Diagnostic Center application demonstrates how ServiceNow custom application development can automate healthcare operations. By integrating table design, dynamic portal workflows, rich-text report generation, automated email notifications, and Platform Analytics dashboards, the solution minimizes administrative overhead and improves patient satisfaction.

8.4 Conclusion

The successful implementation of the Medical Diagnostic Center platform provides a complete solution for managing diagnostic workflows. The application accelerates report delivery, eliminates scheduling errors, maintains transparent transaction logs, and delivers actionable insights to healthcare administrators.